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ORIGINAL ARTICLE

A decoupled time-optimized multi-robot path parameterization method

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Abstract

This paper presents a decoupled time-optimized path parameterization method for coordinating multiple robots with kinodynamic constraints in shared workspaces, where each robot follows a predefined geometric path. The proposed approach first computes time-optimal trajectories for individual robots using the Time-Optimal Path Parameterization (TOPP) algorithm, then resolves inter-robot collisions through strategically calculated start-time delays. By introducing the concept of Generalized Collision Regions (GCRs) in coordination space and developing an efficient algorithm to compute delay collision intervals, our method guarantees collision-free motion while minimizing the total operation time. The solution is formulated as a convex optimization problem with linear constraints, enabling efficient computation. Experimental validation using both robotic arms and car-like robots demonstrates the method's effectiveness.

Keywords Multi-robot · Decoupled approaches · Path parameterization · Time delays · Generalized collision region

Introduction

As operational problems become increasingly complex, using multiple robots to work together has become a more viable option. However, coordinating multi-robot effectively when performing independent tasks in the same workspace and avoiding collisions between them remains common major challenges in multi-robot applications.

The two main approaches to multi-robot motion planning are the centralized and decoupled methods [1]. The centralized approach [2] plans trajectories in a high-dimensional configuration space that encompasses all robots. While this method is complete, it becomes difficult to apply with many robots or high degrees of freedom (DOFs). The decoupled approach [3] divides multi-robot motion planning into two steps: path planning and path parameterization. Path planning provides each robot with a collision-free path around static obstacles, and path parameterization plans the velocity profile for each path to prevent collision between robots. Path planning is an established task, and our research focuses on parameterizing the paths of each robot to enable collision-free motion and optimize motion duration as much as possible.

The problem of path parameterization for a single robot has been extensively investigated, with two primary methods employed: numerical convex optimization (CO) and numerical integration (NI) [4]. The CO-based method, introduced by [5], formulates and solves the path parameterization problem as a single large convex optimization program. However, this approach involves a considerable number of variables and constraint inequalities, and its implementation speed is significantly slower than the NI-based algorithm [6], particularly when the discretization step size increases. The NI-based method, proposed by [7], leverages the Pontryagin maximum principle, which dictates that the time-optimal path parameterization consists of alternating maximum acceleration and deceleration segments. The key advantage of this method is its ability to explicitly calculate the optimal control at each point along the path, allowing for

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rapid implementation. Reference [8] introduced the Time-Optimal Path Parameter (TOPP) algorithm, which defines that each joint of the robot satisfies torque constraints while maximizing the speed at each point on the path. Reference [9] further extended this method by incorporating joint velocity constraints and proved its time optimality.

For multi-robot, the coordination space is commonly used for multi-robot path parameterization. In [10], the concept of coordination space was proposed to describe the combinatorial evolution of robot motion and collision region. The coordination planning of two robots in the coordination space has been extensively studied. Several strategies have been proposed in the literature [11–14] to plan collision-free coordination curves within the coordination space, effectively applied to the motion planning of two robots. For the coordination planning of multi-robot within the coordination space, author of [15] employed the D* algorithm to plan the coordination curve and addresses the time optimization motion planning problem for mobile robots. Reference [16] utilized an improved A* algorithm to plan a coordination curve in a three-dimensional (3D) coordination space that avoids collision areas, achieving collision-free operation of three robotic arms. Similarly, researchers in [17] proposed an online coordination method for multi-robot movements, adjusting the time evolution of each robot along its original geometric path based on the movements of other robots, achieving collision-free execution of their respective tasks. The aforementioned methods aim to identify the shortest coordination curve within the coordination space, connecting the initial and final points while circumventing collision zones. However, these approaches do not consider robot kinodynamic constraints during the coordination curve planning process. This omission leads to two critical issues: (1) planned coordination curves may require dynamically infeasible motions (e.g., instantaneous velocity changes) when executed by physical robots, and (2) time-optimal trajectories satisfying kinodynamic constraints may violate collision conditions if not properly coordinated. Consequently, the simultaneous consideration of both the coordination curve and kinodynamic constraints becomes imperative - the former ensures inter-robot collision avoidance through temporal synchronization, while the latter guarantees the physical realizability of each individual trajectory. This dual requirement introduces significant challenges to the path parameterization of a single robot, as the solution must satisfy both collaborative and dynamic feasibility constraints. Notably, learning-based approaches such as [18] have demonstrated that locally weighted learning with genetic optimization can complement traditional NI/CO methods by providing adaptive velocity profiles for complex dynamics systems. Reference [19] introduced a multi-robot path parameterization method that does not rely on the coordination space. This method enables the time-optimal planning of multi-robot motion trajectories while adhering to kinodynamic constraints. However, it is only applicable to mobile robots and does not accommodate robots with multiple joints.

In the context of multi-robot coordination with kinodynamic constraints along predefined paths, ensuring collision-free trajectories remains a critical challenge. A widely adopted approach in the literature involves formulating the problem as a Mixed-Integer Linear Program (MILP), which enables the simultaneous optimization of discrete collision-avoidance constraints and continuous kinodynamic feasibility. MILP-based methods, such as those proposed by [19] and [20], leverage binary variables to encode pairwise collision conditions while incorporating linearized dynamics for real-time solvability. Recent research [21] focus on enhancing MILP scalability for large-scale road networks through techniques like binary variable resolution reduction, predetermined vehicle order constraints, and iterative lazy constraint algorithms, significantly improving computational efficiency while maintaining near-optimal solution quality. Although MILP-based methods are theoretically optimal, their handling of collision zones tends to be overly rigid, preventing them from achieving the true optimal solution.

To address the limitations, this paper makes three key contributions to multi-robot motion planning: (1) We propose a novel decoupled path parameterization method that combines time-optimal single-robot trajectories with delay-based coordination, providing an efficient solution for collision-free multi-robot coordination along specified paths; (2) We introduce the concept of Generalized Collision Region (GCR) with a computationally efficient modeling method that extends traditional collision regions to include dead zones and unreachable areas; (3) We develop a delay collision interval calculation algorithm that enables optimal time coordination while satisfying kinodynamic constraints, outperforming existing MILP-based methods in both computational efficiency and solution quality. The proposed method has been validated through simulations with both robotic arms and mobile robots.

Problem modeling

To translate the collision avoidance challenge into a solvable optimization problem, we first formalize the multi-robot coordination framework. The key components include: (1) the coordination space (CS), which encodes the relative progress of robots along their paths; (2) the coordination curve (CC), representing temporal synchronization; and (3) the Generalized Collision Region (GCR), extending traditional collision models to account for kinematic constraints. These constructs enable a systematic decomposition of the high-dimensional planning problem into tractable sub-tasks, as detailed below.

Coordination space

Considering n robots R_i, $i \in \{1, ..., n\}$, moving along specified paths $path_i$, the $path_i$ is a sequence of configurations q_i; $path_i$ is parameterized by path coordinate s_i, where s_i represents the length of the robot's movement along the path and satisfies $0 \leq s_i \leq s_{i\,\max}$. The CS is an n-dimensional space composed of the path coordinate s_i of the n robots.

Coordination curve

The CC is a curve in the coordination space that connects the start point $P_0 = (0, ..., 0)$ and endpoint $P_{end} = (s_{1\,\max}, ..., s_{n\,\max})$ and describes the relative positions of the robots. Any point $(s_1, ..., s_n)$ on the CC satisfies $ds_i/ds_j \geq 0$, $i, j \in \{1, ..., n\}$, which prevents all robots from moving backward along the path. According to [22] the presence of backward motion is always more time-consuming than the absence of backward motion. If the CC does not pass through the collision region, we call it a collision-free coordination curve (CFCC). The goal of multi-robot path parameterization research is to find a suitable CFCC.

Generalized collision region

The path for each robot R_i is specified, and path coordinate s_i uniquely identifies robot configuration q_i of R_i. Thus, each point $P(s_1, ..., s_n)$ in the CS corresponds to the configuration of n robots. If a collision between the configurations of the n robots at a point in the CS, then the point is considered a collision point, and the set of all collision points is called the collision region (CR).

The CR of an n-dimensional CS consists of $C(n, 2)$ cylinders that emanate from the CR of the two-dimensional (2D) coordinate space (Fig. 1), so we discuss the characteristic of the CR in 2D CS. In the 2D CS, the CFCC is not allowed to pass through certain areas, except the CR. We define all areas that the CFCC cannot pass as the GCR. Compared to the CR, GCR encompasses three supplementary regions (Fig. 2): the inner region (IR), the dead region (DR), and the unreachable region (UR). The IR is the area surrounded by the CR, and the CC cannot enter the IR without colliding with the CR. Therefore, while the robots do not collide in the IR, this region cannot be utilized in planning CFCC.

Both the DR and UR are a result of the property $ds_i/ds_j \geq 0$ in CC. As the CC passes through the DR, it is bound to collide with the CR, given the condition of $ds_i/ds_j \geq 0$. Similarly, the CC cannot enter the unreachable zone because of the same property. Based on the aforementioned analysis, we concluded that any point on the boundary of the GCR must satisfy $ds_i/ds_j \geq 0$.

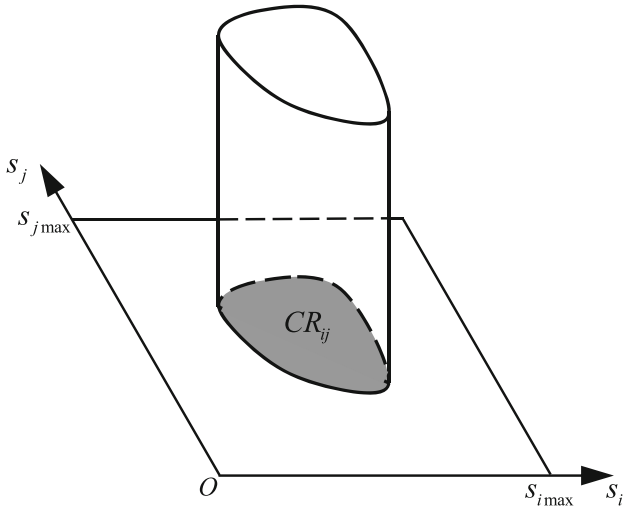


Fig. 1 Collision region cylinder in high-dimensional coordination space

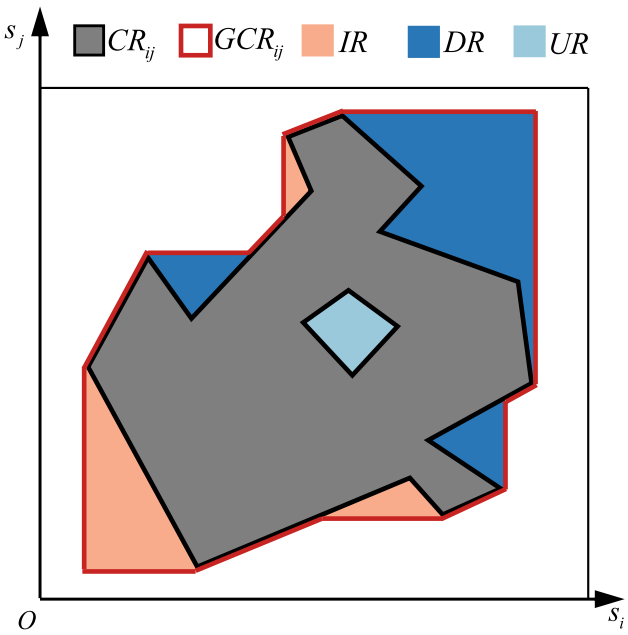


Fig. 2 Comparison between GCR and CR

Time-optimized multi-robot path parameterization method

Single-robot time-optimal path parameterization

The Time-optimal Path Parameterization (TOPP) algorithm for a single robot has been introduced in previous studies [8]. This algorithm operates in a 2D path parameterization space $(s, \dot{s})$, where the path velocity is denoted as $\dot{s}$. The torque constraint of the robot is expressed as an inequality

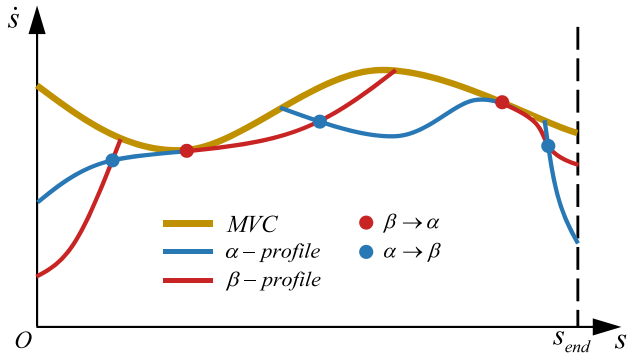


Fig. 3 MVC and α-, β-profiles in TOPP algorithm [8]

that involves s, $\dot{s}$ and $\ddot{s}$.

$$\alpha(s,\dot{s}) \leq \ddot{s} \leq \beta(s,\dot{s})$$

where, $\ddot{s}$ is the path acceleration, while $\alpha(s,\dot{s})$ and $\beta(s,\dot{s})$ denote the minimum path acceleration and maximum path acceleration, respectively. The max velocity curve (MVC) is the boundary of the feasible region for velocity profile planning in the phase plane. The velocity profile must remain below the MVC. The TOPP algorithm achieves time-minimum velocity profile by using bang-bang-type control [23], whereby the optimal profile follows the β and α fileds alternatively while always staying below the MVC(Fig. 3). The TOPP algorithm can make the robot to complete a specific path in the shortest time under the robot kinodynamic constraints.

Multi-robot path coordination

For the multi-robot path parameterization coordination problem, employing time delays to resolve trajectory conflicts among robots is a commonly used method [24, 25]. Reference [25] proposed an approach that combines priority assignment with time delays to address trajectory conflicts between two robots. This method assigns higher priority to one robot and allows the lower-priority robot to delay its departure to avoid collisions with the higher-priority robot.

In the two-dimensional CS, the effect of time delay is manifested in the behavior of the CC. During the initial phase, the CC advances solely along the non-delayed axis (corresponding to the path of the higher-priority robot). Only after the delay period elapses does the CC begin to advance synchronously along both axes toward the endpoint. This mechanism, which shifts the CC positively along the non-delayed axis, ensures that it avoids the CR, thereby achieving collision-free motion between the two robots, as illustrated in Fig. 4.

We propose an extension of the time-delay method to address the multi-robot path coordination problem in high-

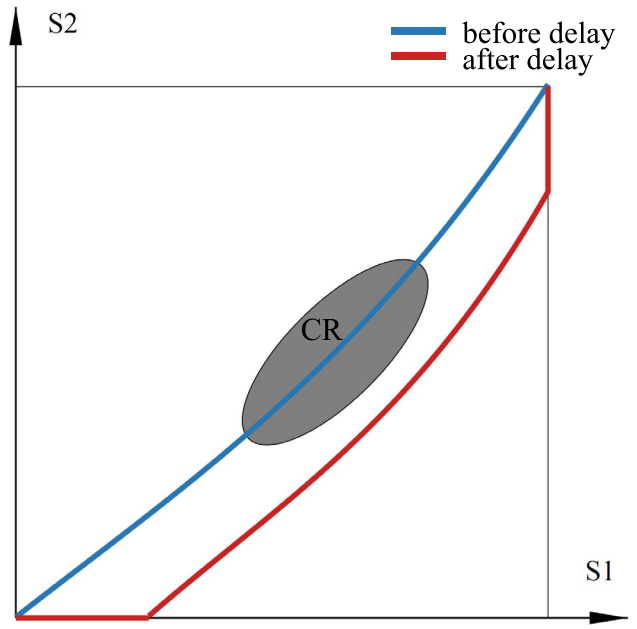


Fig. 4 Two-dimensional coordination space time delay method

dimensional spaces. By adding delays in the start time of specific robots, we effectively resolve the issue arising from conflicting robot trajectories. To illustrate this approach, we consider the CS in three dimensions. In our methodology, time delays are strategically added to the trajectories of robots r_1 and r_3 in the 2D CSs CS_{12}, CS_{13}, and CS_{23} (refer to Fig. 5). This deliberate delay induces a significant shift in the three CCs, which were previously intersecting with the GCR, subsequently redirecting them away from collision paths. As a result, in the 3D CS CS_{123}, the CC that previously caused interference with the GCR no longer intersects with it. Corresponding to the actual operation scenario, the three trajectory interference robots no longer collide after coordination.

For multi-robot coordination, we introduce the concept of the sub-coordination space within the overarching multi-robot coordination framework. Specifically, we define the 2D CS $CS_{ij}(i, j \in \{1, \cdots, n\})$ as a sub-coordination space, which is formed by any pair of robots r_i and r_j.

Property 1: The CC of a sub-coordination space is the projection of the CC in a high-dimensional CS on that 2D plane.

Theorem 1: The necessary and sufficient condition for ensuring that a CC in a multidimensional coordination space avoids collisions with the GCR is that the CC remains free from collisions with GCR_{ij} in all sub-coordination spaces CS_{ij}.

In the sub-coordination space, the introduction of time-delay in robots r_i and r_j causes a shift in the CC towards the positive directions of s_i and s_j, respectively. We define the absolute delay T_{Dij} of the subspace CS_{ij} as the difference

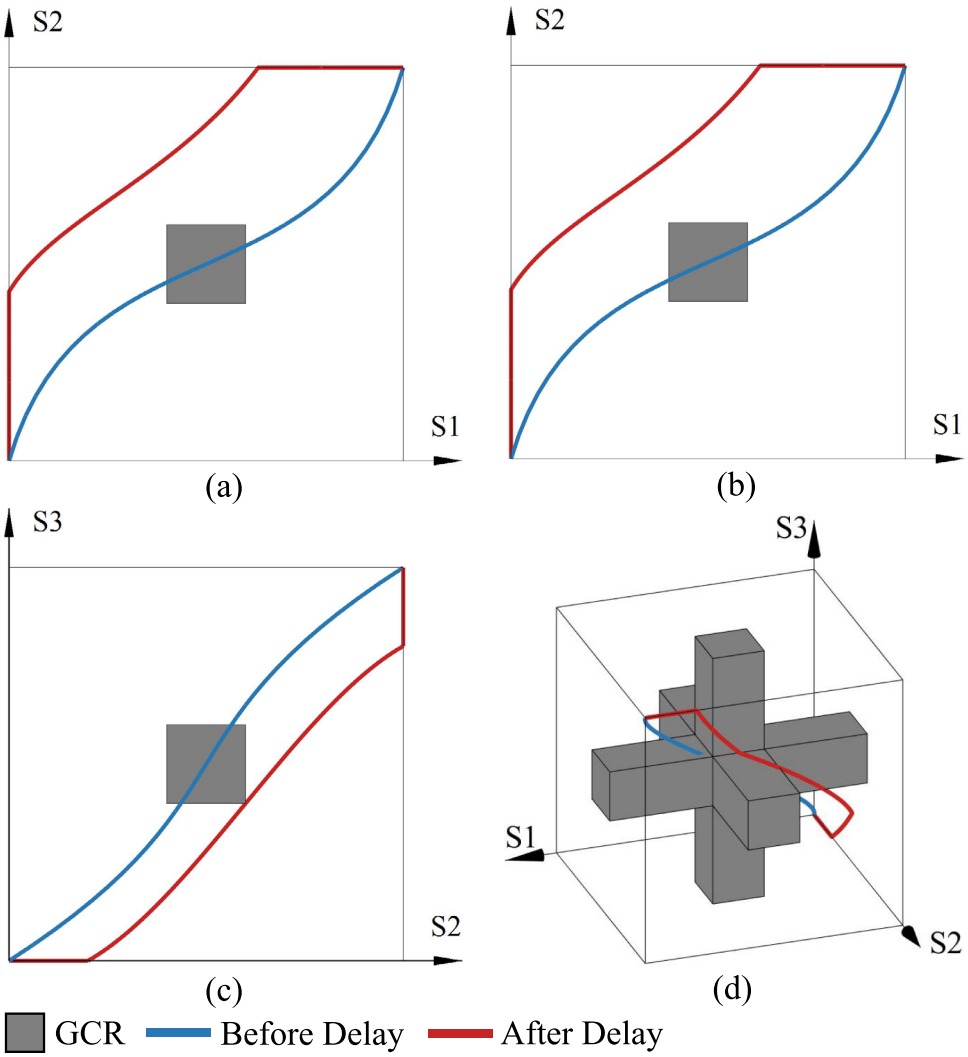


Fig. 5 Three-dimensional coordination space time delay method

between the delay T_{Di} of r_i and the delay T_{Dj} of r_j. The delay collision interval, denoted as $[T_{ij}^{GCR_D}, T_{ij}^{GCR_U}]$, is defined to identify the temporal range where the coordination curve CC_{ij} in the sub-coordination space CS_{ij} intersects with the GCR. Specifically, when $T_{Dij} \in [T_{ij}^{GCR_D}, T_{ij}^{GCR_U}]$, the coordination curve CC_{ij} collides with the generalized collision region GCR_{ij}. Conversely, if $T_{Dij} < T_{ij}^{CR_D}$ or $T_{Dij} > T_{ij}^{CR_U}$, the coordination curve CC_{ij} does not collides with the generalized collision region GCR_{ij}. The operation time, denoted as T_i, of a single robot is determined by the sum of the path duration (T_{Pi}) and the delay duration (T_{Di}), expressed as $T_i = T_{Pi} + T_{Di}$. In the case of multi-robot systems, we define the total time spent on all robot operations as $T_{total} = \max\{T_i\}$.

The objective of multi-robot coordination is to minimize the total operation time. The coordination problem in multi-robot motion planning can be formulated as follows:

$$\begin{aligned} &\text{minimize } \max\{T_{Pi} + T_{Di}\} \\ &\text{subject to } T_{Di} - T_{Dj} \notin [T_{ij}^{CR_D}, T_{ij}^{CR_U}] \end{aligned}$$

where T_{Pi} is a constant (the time-optimal path duration for robot R_i) and T_{Di} is the optimization variable (the delay time for robot R_i).

The convexity of this optimization problem is rigorously established through two key aspects. First, the objective function is convex as it constitutes a pointwise maximum of affine functions ($T_{Pi} + T_{Di}$), where the convexity-preserving property of the maximum operator guarantees overall convexity. Second, each collision avoidance constraint can be decomposed into two linear inequalities: $T_{Di} - T_{Dj} < T_{ij}^{CR_D}$ or $T_{Di} - T_{Dj} > T_{ij}^{CR_U}$, which form convex half-spaces. The intersection of these convex constraints preserves the convexity of the feasible region.

The convexity property ensures that any locally optimal solution is globally optimal, while enabling efficient computation using standard convex optimization techniques. The problem structure guarantees reliable convergence to the optimal solution in polynomial time, making it particularly suitable for real-time multi-robot coordination applications where both solution quality and computational efficiency are critical requirements.

To ensure that the coordination curve avoids collision regions, it is necessary to first accurately model the GCR. The specific method is described as follows.

Modeling method for GCR

In multi-robot motion planning, the CR is a widely used representation for potential collisions between robots [14, 26]. However, as discussed in Generalized collision region, CR exhibits incompleteness (e.g., by neglecting unreachable and dead regions), making it insufficient to guarantee collision-free coordination curves. To address this limitation, we extend CR to the GCR. This section details the methodology for constructing GCR based on CR.

The GCR of an n-dimensional CS is composed of $C(n, 2)$ combinations of GCRs from sub-coordination spaces. The GCR_{ij} of the sub-coordinated space CS_{ij} is an extension based on CR_{ij}. The CR_{ij} is obtained as follows: $path_i$ is uniformly sampled in the range $[0, s_{i\,\max}]$; $path_j$ is uniformly sampled in the range $[0, s_{j\,\max}]$. Collision detection is performed on the robot configurations corresponding to each pair of sampled points, and if a collision is detected, it is included in the CR_{ij} set.

The method to expand CR_{ij} into GCR_{ij} is as follows:

Step 1: We identify the minimum value $s^{CR}_{i_\min}$ and the maximum value $s^{CR}_{i_\max}$ within the CR_{ij} set along the S_i direction. Subsequently, we compute the maximum- and the minimum- value point in the S_j direction for each point within the CR set, ranging from $s^{CR}_{i_\min}$ to $s^{CR}_{i_\max}$. To avoid discontinuities, we check if the difference between the coordinates of adjacent maximum-value points (or minimum-value points) along the S_j direction exceeds the S_j direction's sampling step size, known as $step_j$. In such cases, we introduce additional extension points to bridge the gap, as depicted in Fig. 6 (a). Finally, we assemble maximum value points, minimum value points, coincident points (where maximum and minimum values coincide), and extension points into the candidate boundary point set of GCR_{ij}.

Step 2: We identify the minimum value $s^{CR}_{j_\min}$ and the maximum value $s^{CR}_{j_\max}$ within the CR_{ij} set along the S_j direction. Subsequently, we compute the maximum- and the minimum- value point in the S_i direction for each point within the CR set, ranging from $s^{CR}_{j_\min}$ to $s^{CR}_{j_\max}$. To avoid discontinuities, we check if the difference between the coordinates of adjacent maximum-value points (or minimum-value points) along the S_i direction exceeds the S_i direction's sampling step size, known as $step_i$. In such cases, we introduce additional extension points to bridge the gap, as depicted in Fig. 6 (b). Finally, we assemble maximum value points, minimum value points, coincident points (where maximum and minimum values coincide), and extension points into the candidate boundary point set of GCR_{ij}.

Step 3: In the CR_{ij} set, we perform a search to identify points $\left(s^{CR}_{i_i\,\min}, s^{CR}_{j_i\,\min}\right)$, $\left(s^{CR}_{i_i\,\max}, s^{CR}_{j_i\,\max}\right)$, $\left(s^{CR}_{i_j\,\min}, s^{CR}_{j_j\,\min}\right)$, and $\left(s^{CR}_{i_j\,\max}, s^{CR}_{j_j\,\max}\right)$ that correspond to the minimum and maximum values along the directions of S_i and S_j, respectively, in Fig. 6c. Subsequently, we enhance the candidate boundary point set of GCR_{ij} by including the boundary points situated between $\left(s^{CR}_{i_i\,\min}, s^{CR}_{j_i\,\min}\right)$ and $\left(s^{CR}_{i_j\,\min}, s^{CR}_{j_j\,\min}\right)$ and between $\left(s^{CR}_{i_j\,\min}, s^{CR}_{j_j\,\min}\right)$ and $\left(s^{CR}_{i_j\,\max}, s^{CR}_{j_j\,\max}\right)$. These boundary points are indicative of the right-angle boundaries present within the lower-left and upper-right regions of CR_{ij}. Notably, these two regions are categorized as the DR and UR, respectively.

Step 4: Internal points within the candidate boundary point set of GCR_{ij} are eliminated, preserving solely the boundary points, which constitute the boundary of GCR_{ij} in the 2D CS CS_{ij} Fig. 6d.

In cases where collision points are on the boundary of the CS in the CR_{ij} point set, the area between the boundary and the CR_{ij} point set must be the DR or UR. Therefore, the boundary of the coordination space needs to be combined to expand the CR_{ij} to obtain GCR_{ij}.

Method of delay collision interval calculation

With the GCR boundary established, the next step is to determine the time delays required to shift the coordination curve outside the GCR. This is achieved by calculating the delay collision interval, as detailed in Algorithm 1.

Algorithm 1 describes the computation process for delay collision interval. For any given point $\left(s^{GCR}_{i_k}, s^{GCR}_{j_k}\right)$ where $k \in (0, ..., size(\partial GCR_{ij}))$, located on the GCR boundary, the TOPP algorithm is employed to calculate the time T_{i_k} it takes for robot r_i to reach $s^{GCR}_{i_k}$ and the time T_{j_k} for robot r_j to reach $s^{GCR}_{j_k}$. For the time difference $\Delta T_k = T_{i_k} - T_{j_k}$, a set of time differences ΔT is obtained by traversing the boundary points of GCR_{ij}. The lower bound of the delay collision interval is the minimum value in the ΔT, and the upper bound is the maximum value in ΔT. When the GCR_{ij} boundary intersects with the boundary of the coordination space CS_{ij}, adjustments need to be made to the delay collision interval. If the GCR_{ij} boundary intersects with the boundaries $s_i = 0$ and $s_j = s_{j_\max}$, then the lower bound

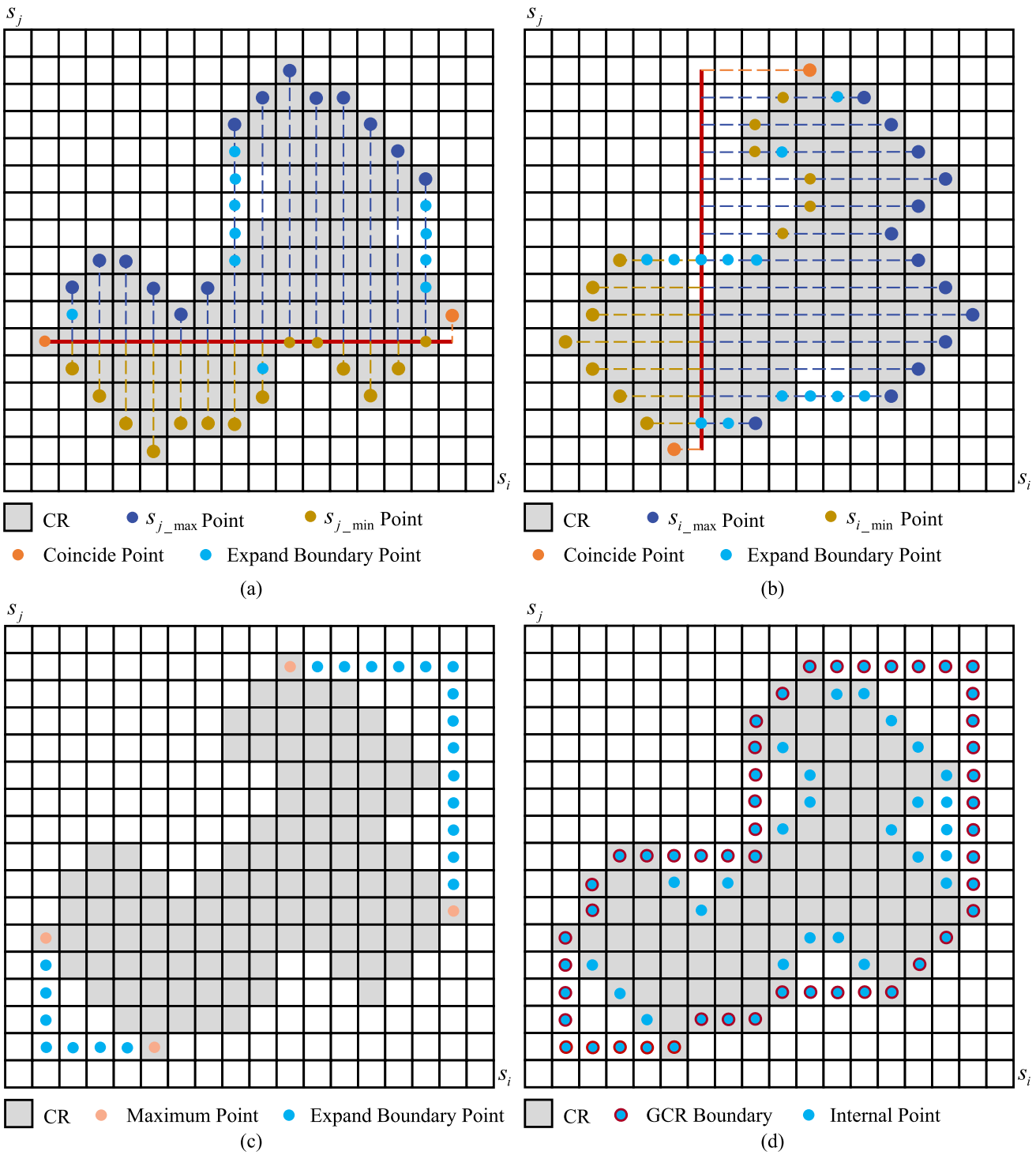


Fig. 6 Modeling method for GCR

Algorithm 1: Calculate the delay collision interval

Input: The trajectories $Profile_i$ and $Profile_j$ of robot r_i and robot r_j obtained using TOPP algorithm, ∂GCR_{ij}(boundary of GCR_{ij}) and ∂CS_{ij}(boundary of CS_{ij})

Output: $\left[T_{ij}^{GCR_D}, T_{ij}^{GCR_U}\right]$

```
for k ← 0 to size(∂GCR_ij) do
    ΔT_k = Profile_i^{-1}(s_{i_k}^{GCR}) − Profile_j^{-1}(s_{j_k}^{GCR});
    ΔT = {ΔT, ΔT_k};
end
T_ij^{GCR_D} = min{ΔT}, T_ij^{GCR_U} = max{ΔT};
while ∂GCR_ij ∩ ∂CS_ij = ∅ do
    switch the boundary intersecting with ∂GCR_ij in ∂CS_ij do
        case s_i = 0
            T_ij^{GCR_D} = −∞, break
        case s_i = s_{i_max}
            T_ij^{GCR_U} = +∞, break
        case s_j = 0
            T_ij^{GCR_U} = +∞, break
        case s_j = s_{j_max}
            T_ij^{GCR_D} = −∞, break

    endsw
end
```

of the delay collision interval is changed to $-\infty$. If the GCR_{ij} boundary intersects with the boundaries $s_j = 0$ and $s_i = s_{i_\max}$, then the lower bound of the delay collision interval is changed to $+\infty$.

Computational complexity analysis

The proposed multi-robot path parameterization method consists of four key components: (1) Single-Robot TOPP, (2) GCR Modeling, (3) Delay Collision Interval Calculation, and (4) Delay Calculation. The computational complexity of each component is analyzed as follows. For the single-robot trajectory optimization using the TOPP algorithm, the computation requires $O(mN)$ operations for each robot, where m represents the number of constraint inequalities and N is the path discretization grid size [4]. Since the TOPP computation is performed independently for each of the n robots, the total complexity for this stage is $O(nmN)$.

The GCR modeling phase involves constructing the generalized collision region by analyzing pairwise sub-coordination spaces CS_{ij} for all robot pairs $\left(R_i, R_j\right)$. The collision detection between $N \times N$ sampled configurations along their paths requires $O(n^2)$ operations per pair. With $C(n, 2) = n(n-1)/2$ possible pairs, the collision detection complexity becomes $O(n^2N^2)$. The boundary extraction from K collision points in each CS_{ij} space has complexity $O(K)$. Considering M ($M \leq C(n, 2)$) actual colliding robot pairs, the total complexity for GCR modeling sums to $O(n^2N^2) + O(MK)$.

In the delay collision interval calculation stage, the TOPP algorithm is reused for each boundary point of GCR_{ij} to compute arrival times T_{i_k} and T_{j_k}. As the number of boundary points scales with the path discretization N, this step requires $O(N^2)$ time per colliding pair, resulting in an overall complexity of $O(MN^2)$ for all pairs.

The final delay calculation solves a convex optimization problem with n variables (delays T_{Di}) and M linear constraints. Using interior-point methods, this problem can be solved in $O(\sqrt{M}n^2)$ time.

This comprehensive complexity analysis demonstrates that the proposed method maintains polynomial-time scalability with respect to the number of robots and path discretization resolution. The modular structure of the algorithm allows for potential optimizations in each component, making it suitable for real-time implementation in practical multi-robot systems with moderate scale. The worst-case complexity remains bounded even in dense collision scenarios, ensuring reliable performance across various operational conditions.

Experimental results

Robot arms experiment

We conducted a comprehensive validation of the proposed algorithm by utilizing three seven-DOF Barrett WAM robots operating within the same workspace. For accuracy and reliability, the simulation was performed on the OpenRave simulation platform [27]. The trajectory planning for the three robots, involving the efficient transfer of objects, was achieved through the implementation of the bi-directional rapidly-exploring random trees (biRRT) algorithm. The parameterization of each planned path was executed using the TOPP library [8], and collision detection between robots was completed using the Flexible Collision Library (FCL).

The experimental setup is depicted in Fig. 7, illustrating three Barrett WAM robots, namely r_1, r_2, and r_3, positioned on the left, right, and middle, respectively. Each robot is tasked with moving the object block from its initial position to its designated endpoint. To achieve this, the paths of the three robots were parameterized using the TOPP algorithm. The resulting trajectory durations for the robots are denoted as follows: $T_1 = 2.0329$ s, $T_2 = 2.0640$ s, and $T_3 = 1.1020$ s. As shown in Fig. 7, prior to coordinating the paths, all three robots experienced collisions with each other.

Fig. 8 exhibits the sub-coordination spaces CS_{12}, CS_{13}, and CS_{23}, along with the 3D CS CS_{123}. In each CS, the original uncoordinated trajectories intersect with the GCR. By employing Algorithm 1, the corresponding delay

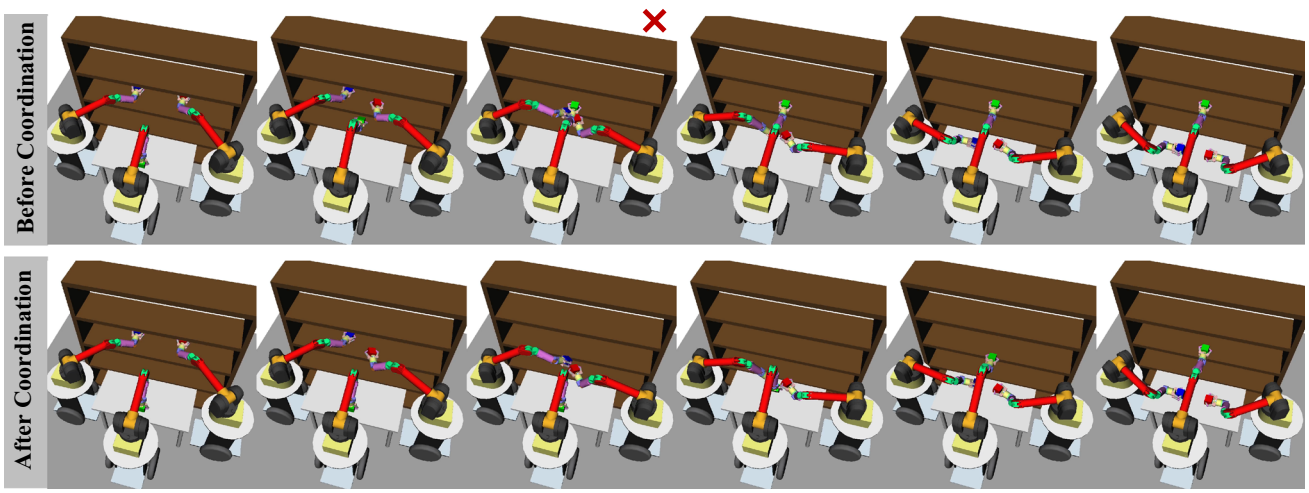


Fig. 7 Comparison of the motion of three robots before and after coordination

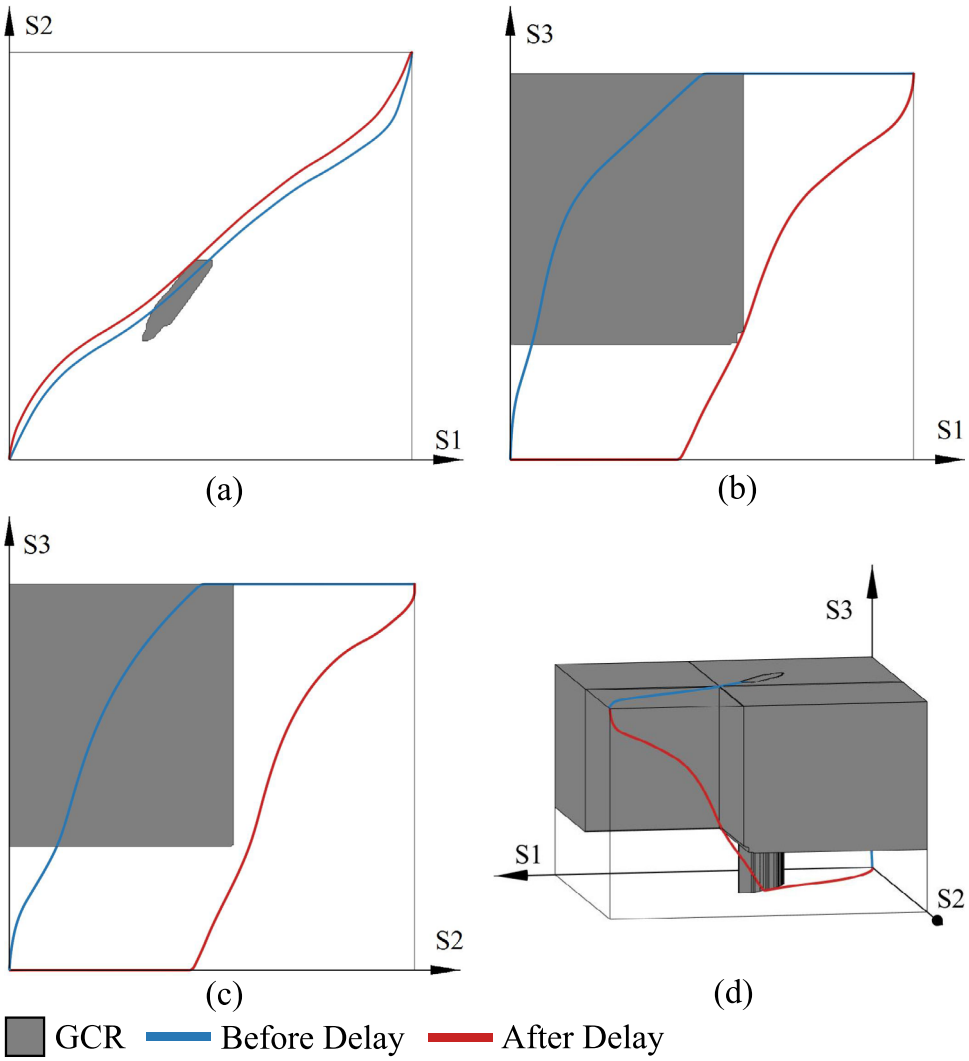


Fig. 8 Coordination curve in coordination space and sub-coordination space before and after coordination

collision intervals for the three CSs were calculated as $[-0.0695, 0.1562]$, $[-\infty, 0.9514]$, and $[-\infty, 0.9346]$. Utilizing the Gekko [28] optimizer, the time delays required for the collision-free paths, considering the time-optimal path, were determined as $T_{D1} = 0.0695$ s, $T_{D2} = 0$ s, and $T_{D3} = 1.0209$ s for robots r_1, r_2, and r_3, respectively. Consequently, the departures of robots r_1, r_2, and r_3 were delayed by T_{D1}, T_{D2}, and T_{D3}, respectively. Following the delays, the resulting motion durations for the three robots are denoted as $T'_1 = 2.1024$ s, $T_2{}' = 2.0640$ s, and $T_3 = 2.1229$ s, respectively. The total duration of the robot operation is denoted as $T = \max\{T_i\} = 2.1229$ s. Subsequent to coordination, the CCs obtained within each CS in Fig. 8 did not intersect the GCR. Additionally, in the simulation environment depicted in Fig. 7, the robot successfully avoided collisions.

Figure 9a–c correspond to the path parameterization results of robots r_1, r_2, and r_3, respectively, showcasing the joint torque variations. The solid lines in the figure illustrate the curves representing the joint torque, while the dashed lines correspond to the torque limits of the respective colored joints (certain joint torque limits are not displayed because they exceeded the drawing range). Notably, the figure vividly demonstrates that throughout the motion, the torque exerted on all joints of the three robots remained within the permissible range of joint torque limits. This is contribution of the TOPP algorithm. Our multi robot path parameterization method does not change the trajectory of the robot during the motion process, but only shifts the trajectory on the time axis, thus ensuring that the robot's motion meets kinodynamic constraints.

Car-like robots experiment

To further validate the effectiveness of our proposed multi-robot path parameterization method, we conducted a comprehensive simulation involving ten car-like robots navigating predefined trajectories (Fig. 10). The trajectories, composed of interconnected arcs and straight segments, were designed to maintain motion smoothness and ensure kinematic feasibility. All robots shared equal path lengths, with uniform dynamic constraints of 10 m/s maximum velocity and 5 m/s^2 maximum acceleration. The simulation environment, implemented in Python, was executed on a high-performance computing platform (Intel Core i7-13700 @2.4 GHz, 16GB RAM, NVIDIA RTX 4090 GPU).

The experimental procedure began with exhaustive path sampling and pairwise collision detection, identifying 15 distinct collision zones in 269.86 s. Each collision zone was subsequently expanded into a GCR using our proposed methodology, a process completed in 0.43 s. This critical step enabled the coordination curves to avoid not only direct collision regions but also dead zones and unreachable areas. For motion planning, we implemented trapezoidal velocity pro-

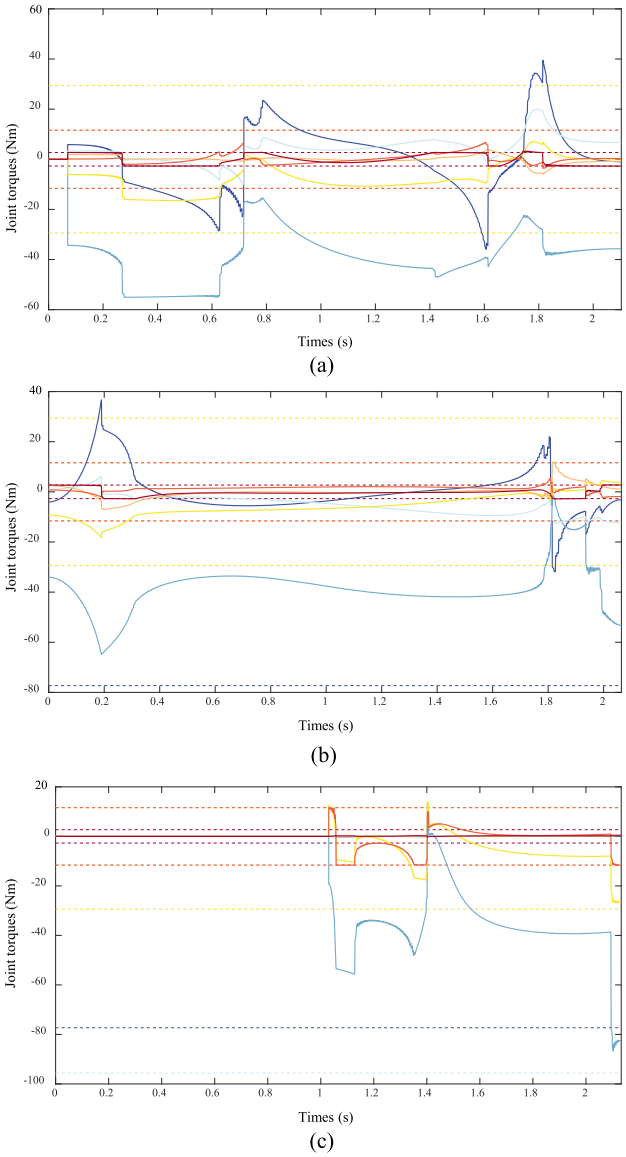


Fig. 9 Torque and joint torque limitations for three seven-DOF Barrett WAM robots after coordination



Fig. 10 Path coordination simulation for 10 car-like robots

Table 1 Comparison of Algorithm Performance Results, the trapezoidal velocity profile represents time-optimal planning without collision consideration

Trapezoidal velocity profile		Our method		MILP-based method[21]	
3pt Collision pair	single-robot traversal time	Computation time	Delay allocation results	Computation time	Additional time consumption
4	21.04s	0.66s	0.25s (robot 4) 1.57s (robot 5) 1.06s (robot 7)	0.74s	1.18s (mean) 2.68s (max)

files where each robot: (1) accelerated maximally to peak velocity, (2) maintained constant velocity, and (3) decelerated maximally to rest. Initial trajectory validation under this profile revealed four residual collisions.

Through Algorithm 1, we calculated precise delay collision intervals for each robot pair, followed by optimal delay time computation using the Gekko optimizer. The experimental results are presented in Table 1. The trapezoidal velocity profile yielded a baseline 20.87s single-robot traversal time without collision considerations. Our coordination strategy introduced strategic delays (0.25s for Robot 4, 1.57s for Robot 5, 1.06s for Robot 7) that eliminated all collisions while minimizing temporal overhead. Comparative analysis with conventional MILP-based approaches revealed significant advantages: where MILP-baesd solutions incurred an average 1.18s (maximum 2.68s) per-robot time penalty beyond baseline, our method achieved collision-free coordination with strictly superior temporal efficiency. This performance gain stems from our precise GCR boundary optimization, which provides additional trajectory planning flexibility compared to MILP-based's restrictive collision-zone exclusion paradigm. Computational overhead remained comparable, with non-collision-detection operations completing in 0.66s (our method) versus 0.74s (MILP-based).

This experiment conclusively demonstrates our method's scalability and efficacy in complex multi-robot scenarios. The synergistic combination of trapezoidal velocity planning and delay-optimized coordination successfully resolved all collisions while strictly adhering to kinematic constraints. These results underscore the method's practical viability for real-world applications requiring coordinated multi-agent systems in shared workspaces, particularly where temporal efficiency is paramount.

Conclusion

This paper presents a decoupled, time-optimized path parameterization method for multi-robot systems operating in shared workspaces, specifically addressing collision-free coordination of multiple robots with kinodynamic constraints along specified paths. The proposed approach combines time-optimal trajectory planning for individual robots with an efficient delay-based coordination strategy, ensuring both collision avoidance and minimal total operation time. Through extensive simulations involving robotic arms and car-like robots, we demonstrate that our method outperforms conventional MILP-based approaches in computational efficiency and temporal optimization. The key innovation lies in the generalized collision region modeling and delay interval calculation, which provide a computationally tractable solution for multi-robot coordination under kinodynamic constraints. Future work will focus on extending the method to handle real-world challenges such as sensor noise and communication delays. These extensions will further enhance the method's applicability in practical robotic systems.

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Data availability The data that support the findings of this study are available from the corresponding author, Liu, G., upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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